DEEP-DIVE ARCHITECTURAL & LOGIC AUDIT #03G: MARTIAL ARTS & TECHNIQUES (PART 7 — FORBIDDEN ARTS, LIFESPAN SACRIFICE, TABOO SEAL ENGINE, SECT FORBIDDEN VAULT & KARMIC LAW ENFORCEMENT)

- **Target Codebase**: `JianghuSect 2` (`JS2-v2.6.0` / `SIM.PATCH_VERSION = '2.6.0'` / `HC.FINGERPRINT = '155:2f9c9dd7'`)
- **Target Audience**: Autonomous AI Software Engineers (**GPT** & **Gemini** Models)
- **Series Context (Completes the 7-Part Martial Arts & Techniques Audit `03A–03G`)**:
  - `03A`: Active Combat Moves (`COMBAT_TECHNIQUE` / `moveDefs`), `SK.list` Caching, `chooseSkill` AI, 16 Weapons & Spatial Range.
  - `03B`: Composable Keyword Engine, 34 Elemental Reactions (`DARK_YIN` vs `YIN_SHADOW`), & Duet Synergies.
  - `03C`: Mantras & Inner Qi Circulation (`MANTRA` / `MANTRA_SYSTEM`), Lifespan Traps & Sect Formation Warfare.
  - `03D`: Qinggong, Sword Flight, Celestial Levitation, Stealth/Recon Arts & Combat Mobility.
  - `03E`: Defensive Kungfu (`DEFENSIVE_ARTS`), Passive Slots (`passiveLoadout`), Innate Arts & Passive Constellations.
  - `03F`: Support Arts (`SUPPORT_ARTS`), Breakthrough Protection, Distress Signals & 64+ Profession/Lifestyle Manuals.
  - **`03G` (This Document)**: Dedicated, exhaustive deep dive into **Forbidden Arts & High-Stakes Taboo Techniques (`FORBIDDEN_ART`, `FORBIDDEN_TECHNIQUE`, `FORBIDDEN_GRAND_DAO`, `CULT.FORBIDDEN_ARTS`, `SECT.FORBIDDEN_VAULT`, and Karmic/Disciplinary Consequences)**.

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0. Hard Architectural Invariants for AI Implementers

1. **Immutable `human_core.js`**: Never modify `GAME_SOURCE/js/human_core.js` (`HC.FINGERPRINT === '155:2f9c9dd7'`). Store permanent lifespan burn or taboo scars in `npc.ext.taboo` / `npc.forbiddenArtState`.
2. **40-Script Load Order & Closed-Loop Economy**: Altar rituals (`AWAKEN_BLOODLINE_ALTAR`, `PURIFY_TABOO_BACKLASH`) must pay real Spirit Stones (crediting `W.world.resourcePool.stones`) and real herbs/beast cores from `W.sect.stock`.

3. **100% Emergent NPC Autonomy**: Temptation to study or unseal Forbidden Arts, fear of backlash, righteous refusal, sect vault policy setting, disciplinary trials, and orthodox-vs-demonic diplomacy must all arise naturally from NPC personalities (`ambition`, `integrity`, `sensationSeeking`, `caution`, `heartDemon`)—never waiting for player UI clicks.
4. **Test-Lock Awareness**: `test/v580_forbidden_arts_and_passive_constellations_suite.js:65-68` asserts `35 FORBIDDEN_ART + 2 FORBIDDEN_TECHNIQUE = 37` in `JD.TECH.techniques` plus `6 FORB_*_T1..T6` manuals. Keep those exact catalog counts intact!

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1. [FORB-LORE-01] 100% Fake Taboo Costs & Unimplemented Archetypes Across 37+ Forbidden Manuals

1.1 Generic `1.65x` + `-14%` Current HP Flattening Erases Every Iconic Taboo Sacrifice (`skills.js:167` & `combat.js:910`)

- **Locations**: `data/data_tech.js` (`ART190_XIAN_073..080`, `ART190_BODY_073..080`, `ART190_GOD_073..080`, `FORB_*`), `js/skills.js:167`, `js/combat.js:388-407, 910`
- **Root Cause**:
  - `SK.profile(t)` (`skills.js:167`) collapses **every** legacy `FORBIDDEN_ART` into `{ arch: 'forbid', mult: 1.65, recoil: 0.14, costPct: 22, cd: 26 }`, and `COMBAT.useSkill` (`combat.js:910`) merely subtracts `14%` of current HP (clamped at `1 HP`) plus `exposed`.
  - Worse, in Patch `v5.8.0` (`CULT.FORBIDDEN_ARTS`), **only 1 out of 6 Taboo Archetypes (`SOUL_DEVOURING_ABYSS`) was given unique combat logic in `combat.js:398`**! Searching `combat.js` for `'LIFESPAN_HEAVEN_DEFIANCE'`, `'CORPSE_PUPPET_NECROMANCY'`, `'MERIDIAN_REVERSE_BURST'`, and `'BLOOD_ESSENCE_IGNITION'` returns **ZERO matches (`0` occurrences!)**!

1.2 Six Explicit Taboo Mechanics Promised in Data & Spec That Are 100% Missing at Runtime

Taboo Archetype / Manual Group	**Representative IDs**	**Explicit Lore / `v5.8.0` Spec Promise**	**Actual Runtime Behavior in `JS2-v2.6.0`**
1. Permanent Lifespan Sacrifice (เผาผลาญอายุขัยจริง)	`FORB_LIFESPAN_BURN_STRIKE`, `ART190_XIAN_075`, `BODY_073/076`, `GOD_073/076/079`, `FORB_LIFESPAN_HEAVEN_DEFIANCE_T5`	Burn `2–15` years of real `lifeMax` (turning hair prematurely white) for armor-piercing star/blood devastation + `<25%` HP Heaven-Defying Last Stand	**`0` years of `lifeMax` ever deducted**; `LIFESPAN_HEAVEN_DEFIANCE` does not exist in `combat.js`!
2. Meridian Rupture & Cultivation Backlash (ชีพจรฉีกขาด/เสียตบะ)	`ART190_XIAN_073/078/080`, `BODY_074`, `GOD_077`, `FORB_MERIDIAN_REVERSE_BURST_T2`	`+30%` walk/Qinggong speed & barrier-shattering burst, leaving post-fight ruptured-meridian injury & risking `cult.prog` loss	**`0%` walk/Qinggong bonus** (`ai.js` never checks `MERIDIAN_REVERSE_BURST`); **`0` post-fight injuries**!
3. Corpse-Puppet & Parasitic Blood-Gu (ปลุกศพหุ่นเชิด/กู่โลหิต)	`FORB_CORPSE_PUPPET_ARMY`, `ART190_XIAN_076`, `FORB_CORPSE_GU_NECROMANCY_T3`	Summon fearless Corpse-Puppets / parasitic Blood-Gu phantoms; inflict uncurable Gu corrosion & drain target stamina	**`0` summons spawned** (`arch` is `'forbid'`, not `'summon'`); **`0` stamina drain**!
4. Soul-Devouring Cultivation Harvest (สูบตันเถียน/เก็บเกี่ยววิญญาณ)	`FORB_BLOOD_SOUL_DEVOUR`, `ART190_XIAN_074`, `FORB_SOUL_DEVOURING_ABYSS_T4`	Siphon HP/Qi on hit **and harvest `+5.0` cultivation progress (`cult.prog`) per slain enemy** (`ACTIVE_PATCH_PROMPT_V5_8_0.md:25`)	Siphons HP/Qi (ignoring `healBlock`!), **never grants `+5.0` kill harvest** (`C.slay` only checks `path === 'DEMON'`)!

Taboo Archetype / Manual Group	Representative IDs	Explicit Lore / `v5.8.0` Spec Promise	Actual Runtime Behavior in `JS2-v2.6.0`
5. Forced Beast-Soul Sacrifice (บังคับอสูรคู่สัญญาเผาชีวิตสู้)	`ART190_BODY_080`, `ART190_GOD_075`	Overdrive bonded Spirit Beast for massive joint strike at the cost of beast HP & `bond` trust loss	**Never reads or modifies `a.pets` (`0%` effect)**!
6. Emergency Dimensional Tear Escape & Faith Harvest	`ART190_XIAN_078`, `GOD_079` (Escape); `FORB_FAKE_FAITH_HARVEST` (Faith drain)	Tear dimensional locks to escape lethal raids/ambushes at heavy meridian cost; or drain sect reputation/believers to rush cultivation	Neither escapes combat/expeditions nor interacts with `W.sect.rep` / faith!

1.3 Remediation Blueprint (`[FORB-LORE-01]`)

- **Real Lifespan Burn (`n.ext.burnedLifespanYears`)**: Subtract `n.ext.burnedLifespanYears` inside `CULT.lifeMax(n)`! When an NPC casts a Lifespan-Sacrifice Forbidden Art in a **lethal fight (`f.serious === true`)**, deduct `1.5–4.0` real years of `lifeMax` (`burnedLifespanYears += costYears`), set `n.ext.whiteHairFromTaboo = true` (rendered in `Render.diorama` / `drawNpc`!), log a dramatic Chronicle event (`"ยอมเผาผลาญอายุขัยฝืนชะตาฟ้า"`), and deliver true shield-piercing devastation (`penAdd +45`)! (Refuse cast if remaining lifespan `< 5` years unless protecting a `companionId`/`disciple` to the death!).
- **Wire All 6 Archetypes Distinctly**:
  - `CORPSE_PUPPET_NECROMANCY` / `FORB_CORPSE_PUPPET_ARMY`: Spawn `1–2` temporary `Corpse-Puppet` summons on cast/kill + apply `crimsonGuSwarm` & `-6` target `energy`.
  - `MERIDIAN_REVERSE_BURST`: Add `+25%` speed in `AI.walkSpeed` / combat stride while unsealed, and apply a 1-day ชีพจรย้อนกระแส (`sev: 1`) recovery debuff when unsealing ends.
  - `SOUL_DEVOURING_ABYSS`: Grant `+5.0` `CULT.gain` in `C.slay` on lethal kills (`f.serious`), accompanied by `+2.5 heartDemon` and `-12 virtueKarma` (`+12 sinKarma`).
  - `ART190_BODY_080` / `GOD_075`: Require an active pet in `a.pets`, multiply strike by `1.45×`, and apply `fatigue += 35, bond -= 8` on the pet!

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2. [FORB-VAULT-02] SECT.FORBIDDEN_VAULT Policies (STRICT_ORTHODOX_BAN) Have Zero Runtime Enforcement!

2.1 Repository-Wide Proof: W.sect.forbiddenVault.policy Is Never Checked by Any Gameplay Module

- **Verified Fact**: Searching the entire codebase for forbiddenVault and STRICT_ORTHODOX_BAN returns **only cultivation.js (setter), ui.js, render.js, v580 test, and markdown reports—zero checks exist in cpeco.js, martial.js, cognition.js, ai.js, combat.js, or sect.js (discipline / censor / diplomacy)**!
- **Logical & Contextual Absurdities**:
    1. **Openly Studying Necromancy Under STRICT_ORTHODOX_BAN**: Even when the Sect Vault is set to **STRICT_ORTHODOX_BAN (**ผนึกตายห้ามฝึกเด็ดขาด**)**, Stage-1..5 disciples freely buy and study Forbidden Arts in the public Scripture Pavilion with **zero access block**, **zero secrecy**, and **zero punishment from the Discipline Hall (discipline)**!
    2. **Righteous Monks Enthusiastically Study Soul-Devouring Taboos**: COG.techPreference (js/cognition.js) applies **zero moral/alignment filter** on FORBIDDEN_ART! A BUDDHA_DHARMA monk or upright XIAN elder (integrity = 92, compassion = 88) evaluates คัมภีร์มารสูบโลหิตกลืนวิญญาณ purely as a high-power manual!
    3. **Zero Jianghu Diplomatic Reaction to UNRESTRICTED_BLOOD_CULT & Blood Altar**: Running an open Blood-Cult sect and performing AWAKEN_BLOODLINE_ALTAR blood rituals causes **zero reputation shift** among Orthodox vs Demonic rival sects in W.world.sects!

2.2 Remediation Blueprint ([FORB-VAULT-02] — Emergent Law, Temptation & Secret Study)

2. **Enforce forbiddenVault.policy With Authentic Human Temptation (cpe + cognition + discipline)**:
    - **STRICT_ORTHODOX_BAN**: Block normal public CPE study of FORBIDDEN_ART manuals (W.sect.library moves them into W.sect.forbiddenVault.sealedManuals, grants +0.2/day orthodox fame, and calms sect-wide heartDemon). **However**—preserving true emergent drama!—a disciple with **high ambition >= 78, greed >= 75, heartDemon >= 45, or BLACK_THREAD vengeance** (or holding a personal n.inv.techs / crystals taboo scroll!) can **secretly practice the Forbidden Art at night (steal/secret_study)**, risking detection by guard patrols and the discipline / censor officers!

- **Disciplinary Consequence When Caught**: Under `STRICT_ORTHODOX_BAN`, if the `discipline` officer catches a taboo practitioner (or witnesses them unsealing `BLOOD_TABOO_UNLEASHED` inside the sect!), they convene enforcement based on `strictness`: confiscate scroll, imprison (`jail`), cripple/seal taboo state, or trigger a dramatic **Defection Escape** (`resolveDefectorUltimatum` $\to$ joins a Demonic rival sect!).
- **`ELDER_LICENSED_ACCESS`**: Allow legal study/unsealing only for `n.rank >= 4` (Core Disciples, Peak Lords, Elders, Master) or during active `f.kind === 'raid'` sect defense!
- **`UNRESTRICTED_BLOOD_CULT`**: Open access to all ranks + `+15%` taboo cultivation bonus, while increasing daily `heartDemon` pressure and shifting diplomacy (`-attitude` with Righteous/Orthodox sects, `+attitude` with Demonic sects!).

2. **Moral Personality Weighting in `COG.techPreference`**:
   - Subtract `(integrity - 55) * 0.45 + (compassion - 55) * 0.30` (and `-1000` if holding `VOW_ASCETIC_PRECEPT`) from `FORBIDDEN_ART` study score, while adding `(ambition - 55) * 0.40 + (sensationSeeking - 55) * 0.30 + heartDemon * 0.25`! Righteous heroes will shun demonic taboos naturally; desperate avengers and ambitious schemers will covet them!

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3. `[FORB-CMB-03]` Critical Combat Bugs in `CULT.FORBIDDEN_ARTS` (`combat.js:388–455`)

3.1 Friendly-Spar Self-Mutilation & Tournament Taboo Novas (`!f.serious` Not Checked!)

- **Exact Code (`combat.js:388–405` & `441–454`)**:
  - Neither Attacker Blood-Sacrifice (`line 388`) nor Defender `TABOO_DESPERATION_NOVA` (`line 441`) checks **`f.serious`**!
  - If an NPC has `HALF_UNSEALED_FLOW` or `BLOOD_TABOO_UNLEASHED` active during a nonlethal **`spar`** or **`tourney`** match:
    1. They slash their own veins for `-2.5%..-5% MaxHP` on every hit,
    2. Break their lifelong `VOW_ASCETIC_PRECEPT` (`line 391`) over a friendly sparring match, and
    3. Detonate a cataclysmic Blood-Void `TABOO_DESPERATION_NOVA` (`line 448`) in the middle of a friendly sect tournament!
- **Fix**: Gate **both** blocks strictly behind `if (f && f.serious)`!

3.2 Multi-Hit AoE & Poison DoT Self-Drain Exploit (`combat.js:393–397`)

- **Defect**: Inside `C.applyDamage`, `hpCost` (`fb.effHpCostPct * hpMax`) and `tabooStrain += strainRate` execute on **every single `applyDamage` call** without checking `isDotTick` or per-action ICD:
  - A 3-hit AoE hitting 6 raiders (`18` `applyDamage` calls) drains **`18 × 3.5% = 63%` of the caster's own MaxHP in a single skill cast** and spikes `tabooStrain` `18×`!
  - Every `poison` / `bleed` tick in `C.statusTick` drains the poisoner's own blood from across the map!
- **Fix**: Deduct `hpCost` and increment `tabooStrain` **at most once per attack action** (`a._lastTabooCostTick !== W.t`) and **never** on `isDotTick` / reflection ticks; also multiply `SOUL_DEVOURING_ABYSS` healing (`line 401`) by `C.healMultiplier(a, W.t)`.

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4. `[FORB-STRAIN-04]` Toothless `tabooStrain = 100` Cap, Unautonomous Seal Modes & Twin Blood-Altar Desync

4.1 `tabooStrain` Reaching `100` Lacks Real Qi-Deviation (ธาตุไฟเข้าแทรก) Consequences!

- **Locations**: `js/cultivation.js` (`CULT.FORBIDDEN_ARTS`, `CULT.rate`), `js/sim.js` (`SIM.daily` / `SIM.hourly`)
- **Logic Gap**:
  - Unsealing `BLOOD_TABOO_UNLEASHED` grants massive `+48%..+55%` cultivation speed (`v580` line 102) and `+58%..+72%` attack surge!
  - Yet when `tabooStrain` reaches `100`, nothing forces a catastrophic collapse! Without real danger, leaving `BLOOD_TABOO_UNLEASHED` permanently active is strictly optimal.
- **Fix — Authentic Taboo Strain Thresholds**:
  - `tabooStrain >= 60`: Apply rising `+heartDemon` (`+0.4/hr`), `-12%` `meridianIntegrity` stability, and `-10%` `breakthroughChance` (offsetting the raw defiance bonus if unpurified!).
  - `tabooStrain >= 95`: Trigger **Taboo Backlash Deviation (**ปราณต้องห้ามตีกลับธาตุไฟเข้าแทรก**)**—forcibly collapse `sealMode` to `'SEALED_RESTRAINT'`, reset `tabooStrain` to `40`, inflict a `sev: 2` internal meridian injury (ชีพจรปราณต้องห้ามตีกลับ), deduct `12%` current `cult.prog`, and log a Sect/Chronicle event!

4.2 100% Autonomous Seal-Mode & Officer Vault/Altar AI (Eliminating UI Puppeteering)

- **Individual NPC Seal AI (`CULT.FORBIDDEN_ARTS.evaluateAutonomousSealMode(W, n, f)`)**:
  - **Default (`Peaceful Sect Life`)**: Keep `'SEALED_RESTRAINT'` (จิตตั้งมั่น +8%) for all orthodox/cautious NPCs or whenever `tabooStrain >= 55` or `VOW_ASCETIC_PRECEPT` is active.
  - ** Secret / Ambitious Cultivation (`HALF_UNSEALED_FLOW`)**: NPCs on `DEMON` / `VOID_HERETIC_PACT` / `DEVOURING_FLESH` paths—or ambitious risk-takers (`ambition >= 76 && sensationSeeking >= 65`) when `vaultPolicy !== 'STRICT_ORTHODOX_BAN'` (or secretly at night!) and `tabooStrain < 40`—enter `'HALF_UNSEALED_FLOW'` during seclusion/cultivation, sealing back down at `tabooStrain >= 55` to seek `SUPPORT_ARTS` purification!
  - **Lethal Crisis Unsealing (`BLOOD_TABOO_UNLEASHED`)**: During a **lethal fight (`f.serious === true`)** when `hpFrac(n) <= 0.38` (or defending the Sect Main Hall / a dying `companionId` / `disciple` against a stronger boss `powerB > powerA * 1.25`), NPCs who know a Forbidden Art autonomously shatter the seal into `'BLOOD_TABOO_UNLEASHED'` (`fx: '🩸 ฉีกผนึกวิชาต้องห้ามแลกชีวิต!'`), then automatically re-seal to `'SEALED_RESTRAINT'` in `COMBAT.leave` once the battle ends!
- **Officer Vault & Blood-Altar Unification (`sect.js`)**:
  - Unify `SECT.FORBIDDEN_VAULT` with `SECT.INFERNAL_SHRINE` so Demonic/Pragmatic/Orthodox Sect Masters (`SECT.masterDecision`) autonomously set `vaultPolicy` matching the Sect Charter (`STRICT_ORTHODOX_BAN` for Scholastic/Benevolent/Orthodox masters; `ELDER_LICENSED_ACCESS` for Martial/Meritocracy; `UNRESTRICTED_BLOOD_CULT` for Demonic/Shadow masters) and invoke `PURIFY_TABOO_BACKLASH` (via `healer`/`master` when ≥ 2 elders have `tabooStrain >= 50` and stones > reserve) or `AWAKEN_BLOODLINE_ALTAR` before major wars/raids!
  - Record `-10..20` `karmicLedger` (`sinKarma`) when casting Soul-Devouring / Corpse-Necromancy / Fake-Faith arts so Divine/Buddhist Karmic-Smite techniques (`3E-12`, `ART190_GOD_011/036`) naturally counter taboo practitioners!

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5. Implementation & Verification Checklist for GPT / Gemini (`03G`)

- [ ] **Authentic Taboo Mechanics**: Implement real `burnedLifespanYears` deduction in `CULT.lifeMax(n)` + white-hair visual flag on Lifespan-Burn arts

(FORB_LIFESPAN_HEAVEN_DEFIANCE_T5, FORB_LIFESPAN_BURN_STRIKE, XIAN_075, BODY_073, GOD_073/076); wire CORPSE_PUPPET_NECROMANCY summons/stamina-drain, MERIDIAN_REVERSE_BURST speed + post-unseal strain, SOUL_DEVOURING_ABYSS +5.0 kill harvest in C.slay, and Beast-Sacrifice / Dimensional-Escape arts.

- [ ] **Combat Safety Fixes (combat.js:388–455)**: Require f && f.serious for both Blood-Sacrifice strikes and TABOO_DESPERATION_NOVA (never in spar/tourney!); add 1-per-action ICD and !isDotTick guard on hpCost & tabooStrain; enforce C.healMultiplier on SOUL_DEVOURING_ABYSS.
- [ ] **Enforce FORBIDDEN_VAULT Policies & Karmic Law**: Gate public CPE study under STRICT_ORTHODOX_BAN / ELDER_LICENSED_ACCESS, add secret night-study temptation + discipline hall trials/defections, add moral aversion/ambition weights in COG.techPreference, add sinKarma accrual, and shift rival-sect diplomacy under UNRESTRICTED_BLOOD_CULT.
- [ ] **Taboo Strain Backlash & Full Autonomy**: Trigger Qi-Deviation backlash when tabooStrain >= 95; implement evaluateAutonomousSealMode (crisis unsealing in serious fights + post-fight re-sealing) and Sect Master/Healer autonomous FORBIDDEN_VAULT governance.